
Agriculture and Forestry

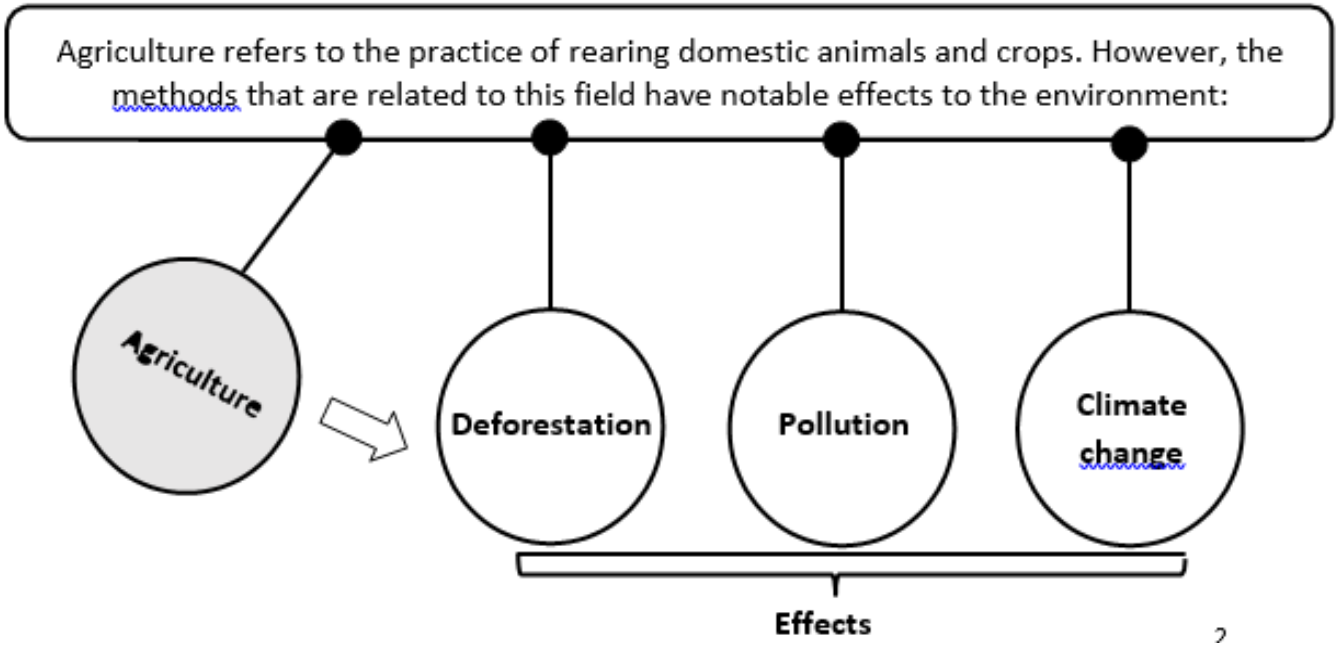
Learning Objectives:



At the end of the lesson the students shall be able to:

- describe the reciprocal relationship between agriculture and climate change; and
- explain the effects of human exploitation on forest resources.

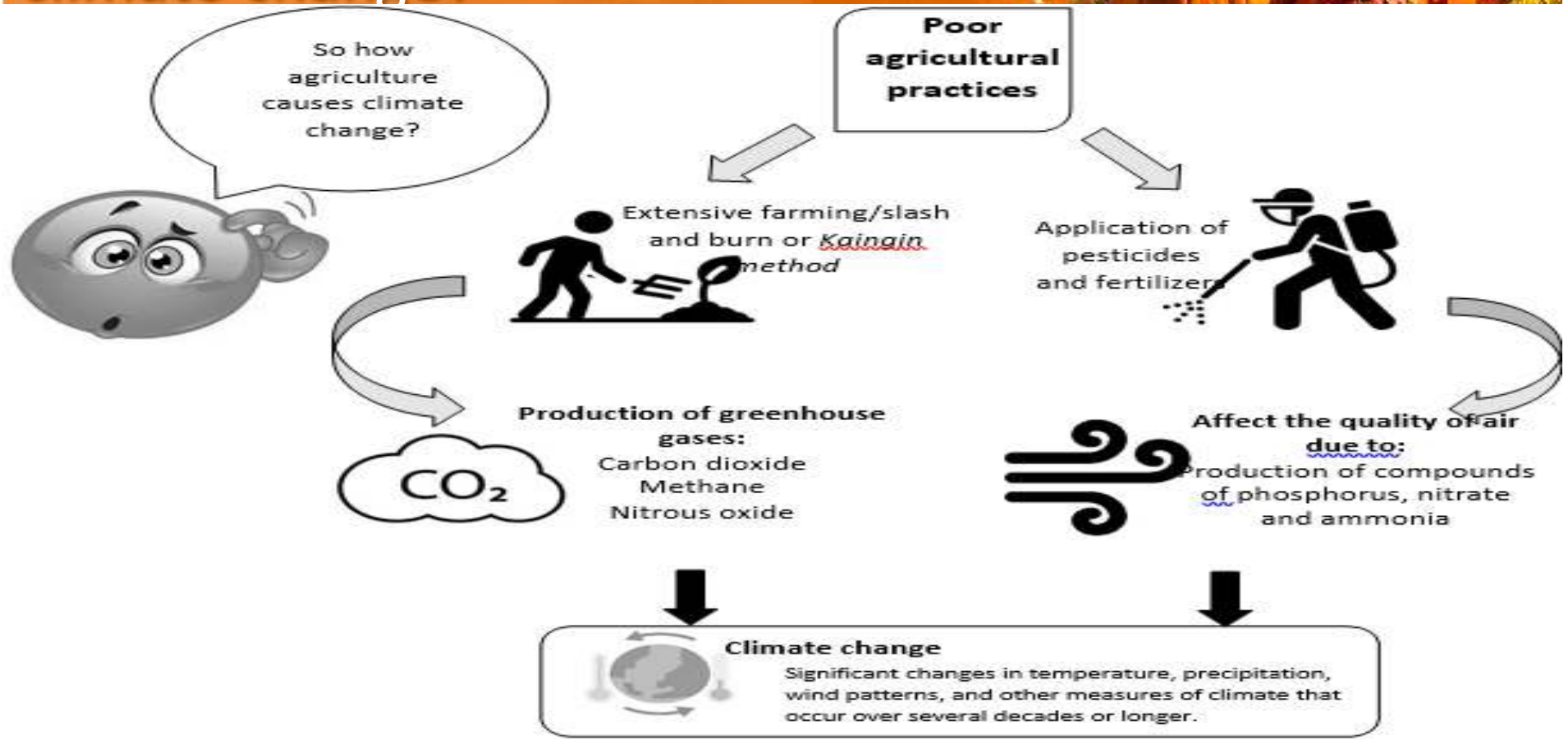
As you can see, humans can both benefit and exploit environmental resources. Now let's discuss deeply on how these fields of agriculture and forestry affect the environment due to human consumption.





How can agriculture results to climate change?





How irresponsible farming can affect the environment:



- Over use of Pesticide and Fertilizers- Once they have been sprayed, it does not disappear completely. Some of it **mixes with the water and seeps into the ground**. The rest is absorbed by the plant itself. As a result, the local streams that are supplied water from the ground become contaminated, as do the animals that eat these crops and plants.



- The use of fertilizers, manure and other organic wastes containing heavy metals such as arsenic, cadmium, mercury and lead can also lead to an accumulation of these heavy metals in the soil.
- When these substances washed into waterways or leach into groundwater sources or get absorbed by plants and are eventually consumed by animals and humans affecting their health or even causing premature deaths. Heavy metals can cause crop failure and poison livestock from contaminated water or food.

How irresponsible farming can affect the environment:



- Contaminated water - This happens due to the disposal of industrial and agricultural waste in local bodies of water.
- Soil erosion and sedimentation – Due to inefficient farming practices, this soil is left open for erosion and leads to declining fertility each year. Whether eroded by water or wind, **all this soil has to be deposited somewhere or the other. The resulting sedimentation causes the soil to build up in areas** such as rivers, streams, ditches and surrounding fields. And so, the process of agricultural pollution **prevents the natural movement of water,** aquatic animals and nutrients to other fertile areas.

How irresponsible farming can affect the environment:



- Livestock - As of now, livestock is grown in cramped conditions where it is fed unnatural diets and sent to slaughterhouses on a regular basis. As a result, they add to the process of agricultural pollution by way of emissions.
- Pests and Weeds - With the arrival of new crops, the native population has to deal with **new diseases, pests, and weeds that it is not capable of fighting**.

How irresponsible farming can affect the environment:



- Excess Nutrients - When excess nutrients are washed into the water systems, e.g., rivers, lakes, streams or oceans during rainy periods, it alters the marine and freshwater nutrient cycles and as an outcome the species composition of the respective ecosystems. The most common consequence is **eutrophication,** which depletes the water dissolved oxygen, and in consequence, can kill fish and other aquatic life.